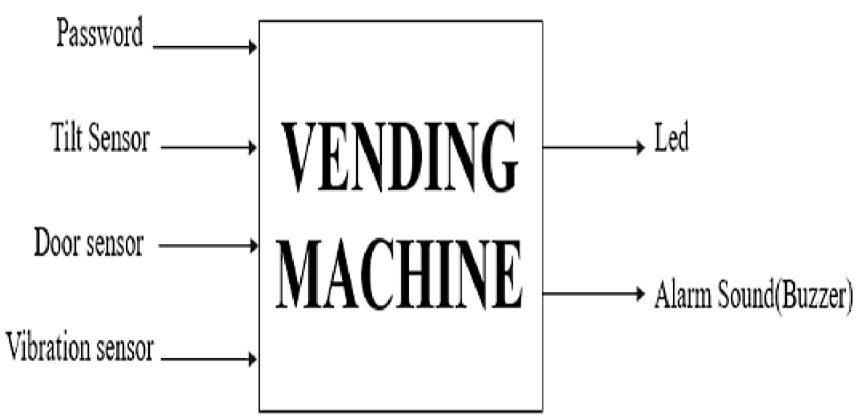
Project Overview

Vending machines are widely deployed in public locations and are often vulnerable to unauthorized access, theft, and physical tampering. This project presents a Verilog-based security system designed to improve the protection of vending machines through password authentication and sensor-based monitoring.

The system continuously monitors various security conditions and activates an alarm whenever unauthorized activity is detected.

The design was implemented using Finite State Machine (FSM) methodology and verified through simulation.

Figure 1: System Block Diagram



Description

The block diagram illustrates the overall architecture of the vending machine security system. Authentication logic, sensors, FSM controller, and alarm modules work together to monitor machine security and respond to violations.

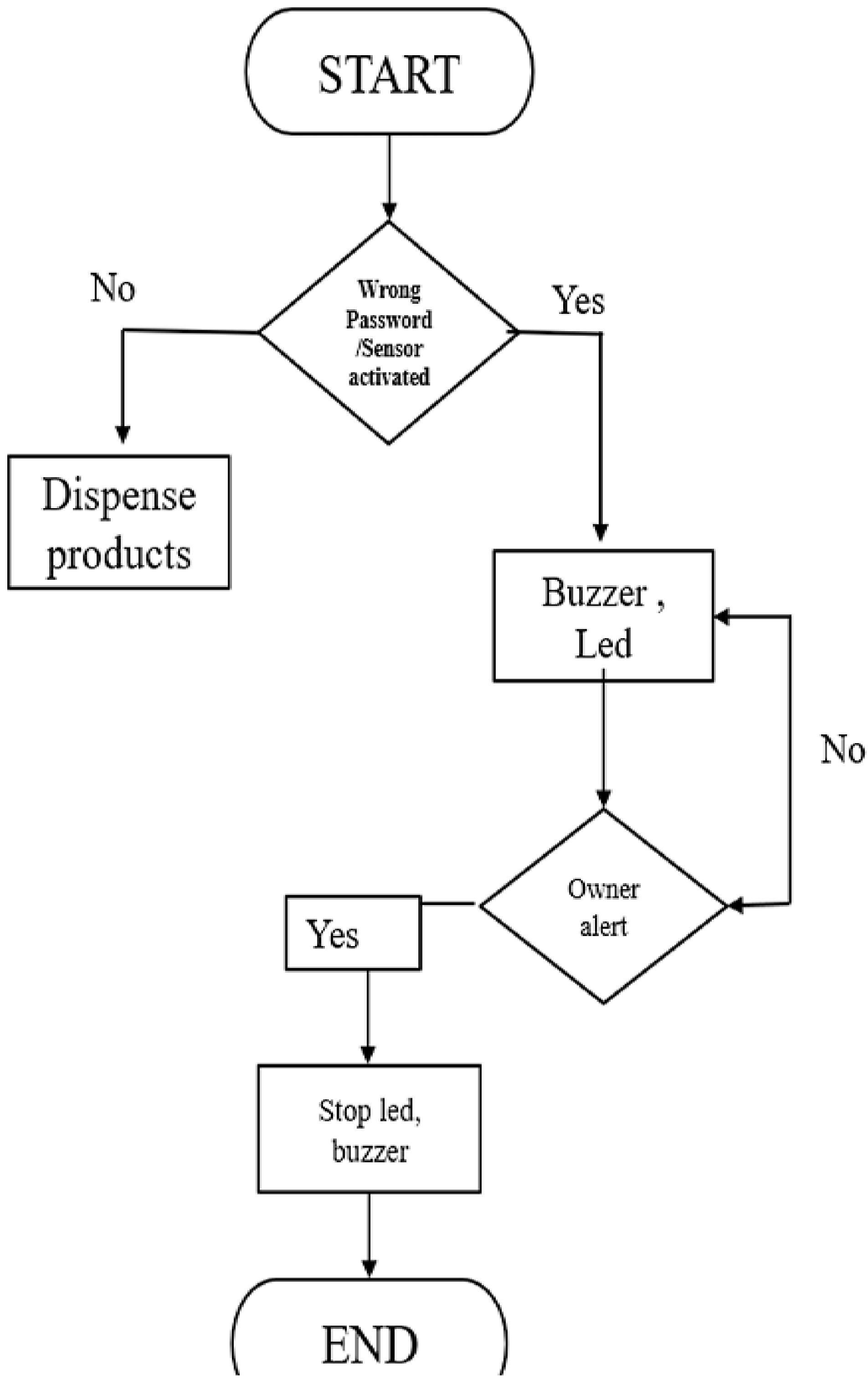
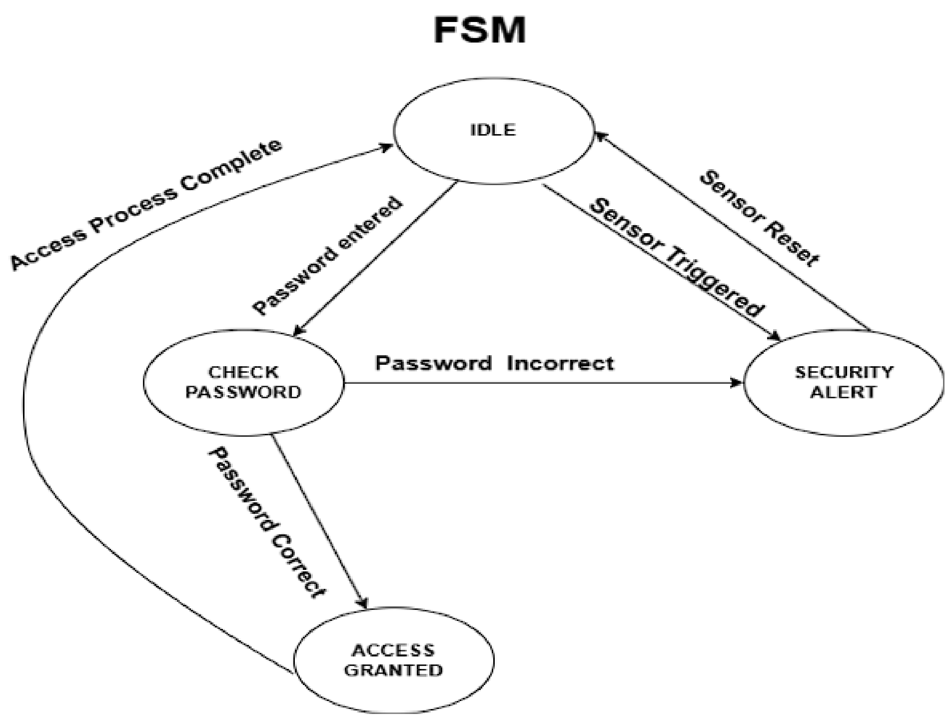


Figure 2: System Flowchart

Description

The flowchart shows the sequence of operations performed by the system. It begins with authentication and continuously monitors sensor inputs before deciding whether to remain in normal operation or activate the alarm.

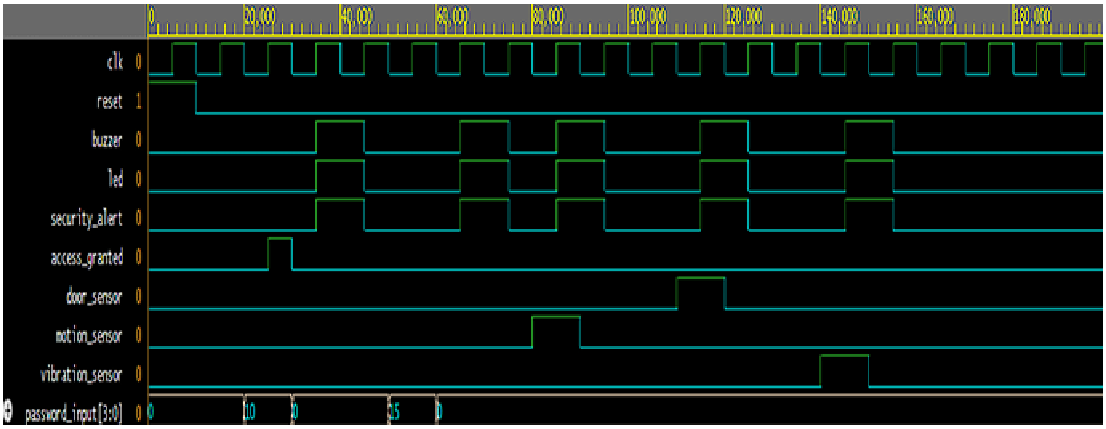
Figure 3: Finite State Machine (FSM)



Description

The FSM controls system behavior by managing transitions between authentication, monitoring, access, and alarm states. State-based implementation improves reliability and simplifies verification.

Figure 4: Simulation Results



Description

The simulation waveform verifies correct system operation. Password validation, state transitions, sensor detection, and alarm activation were successfully tested during simulation.

Observations

- Correct FSM operation
- Successful password authentication
- Accurate sensor response
- Alarm activation during violations
- Stable system performance

Project Outcome

The system successfully demonstrates a Verilog-based security architecture capable of detecting unauthorized access and responding through alarm generation. The FSM-based design provides a reliable and scalable foundation for future smart security applications.